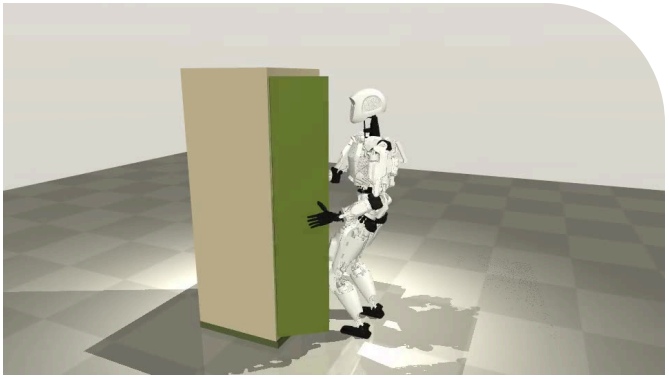
Film a cabinet. Get its URDF. Then a humanoid opens it.

ObjectModel turns a capture of an articulated object into a measured, MuJoCo-validated digital twin. **Twin-to-Task** hands that twin to a humanoid, either the public Apollo model or a generic upper body. The humanoid opens the object in a simulation of its *ground truth*, so the simulator scores whether the twin is good enough to act on.



Independent student project · not affiliated with Apptronik · Apollo model: Google DeepMind MuJoCo Menagerie (Apache-2.0)

Twin accuracy vs humanoid effort

Twin used to plan	Succeed	Median extra force
ObjectModel twin	**100%**	0 N
+2° axis error	60%	16 N
+5°	30%	56 N
+10°	0%	116 N

10 tasks (doors, drawers and lids on 5 objects). Extra force is the grasp force above the same task planned with the true joint. The twins are within about 0.2° of the true axis at 1 mm tracking noise.

Under the hood

- RANSAC rigid segmentation; joint type chosen by BIC; screw-axis fits with bootstrap uncertainty
- Kinematic trees for chains and branches; mass and inertia; URDF and MJCF exports that agree to 0.00 mm
- MuJoCo checks: replay vs capture, collision sweep, stability
- Humanoid: affordances, a stance search over the handle arc, DLS IK on the reference pose (torso and 7-DoF arm on Apollo)
- C++/Eigen core that matches NumPy to 1e-16: revolute fit about 200× faster, 0.16 µs per IK step
- CLI, REST API, MCP server for AI agents, web playground, CI, 130+ tests

Where it meets the roles in Apptronik's public postings

Developer experience	CLI, MCP server for AI coding agents, hosted API, CI, a dev server that mirrors production
Real-time controls	Frames and quaternions, DLS IK with nullspace posture, servo tracking under load, a C++ core with sub-µs IK steps
Motion planning	Upper-body IK on the Apollo model, stance planning, constraint-force metrics
Simulation and sim-to-real	Simulator-in-the-loop validation, parameter estimation from motion, noise and camera-sway sweeps
Autonomy and perception	Perceive → model → plan → act → verify loop; LK tracking, depth lifting, ArUco scale. ROS 2 node is the next step.
Fleet and full-stack	Validated Python serverless API, a static web app with live 3D, deploy config

Run it: `objectmodel task kitchen_cabinet` · `objectmodel task-bench` · `objectmodel task fridge --robot apollo --video fridge.mp4` · Code: github.com/ayanbhardwaj1/objectmodel · Limits: contact-constraint grasp, pelvis held at stand, synthetic captures.